Part 1: Processes, threads, and scheduler primitives

Tests run:

1. **Waitpid tests**: Tests process creation and do_waitpid() behavior for waiting on specific children, any child, invalid PIDs, and multiple child processes.
2. **Context switch test:** Verifies that the scheduler correctly performs context switches between threads while they modify shared data.
3. **Wakeup tests**: Tests threads sleeping on a queue and being awakened using sched_wakeup_on() and sched_broadcast_on().
4. **Uncancellable sleep tests**: Ensures threads sleeping with sched_sleep_on() cannot be cancelled and only wake when explicitly signaled.
5. **Cancellation tests**: Verifies that threads waiting in cancellable sleep can be cancelled correctly in different positions in the wait queue.
6. **Reparenting tests**: Checks that when a parent process exits, its child processes are correctly reparented to the init process.
7. **Race condition and mutex tests**: Demonstrates a race condition on a shared variable and verifies that mutex locks prevent it, including cancellable lock scenarios.
8. **Kill child processes test**: Tests proc_kill() by killing all child processes of the current process while they are sleeping.
9. **Kill all processes test**: Tests proc_kill_all() to terminate all processes except init and ensures the system keeps running.
10. **Sunghan test**: Checks mutex and queue coordination under randomized sleeps

Tests 1 to 9 are part of **faber** tests, **sunghan** is a separate test.

Outputs:

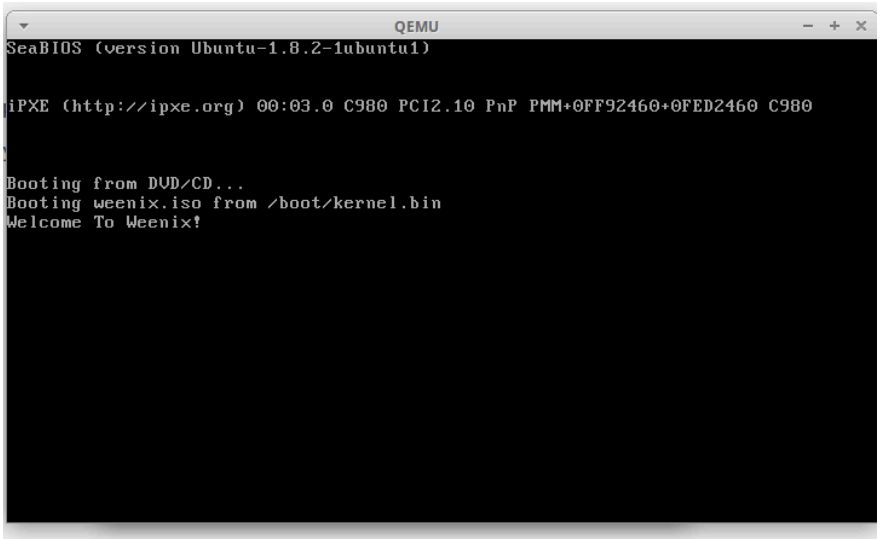
1. Run make clean to remove the existing binary, then run make to build a new binary (I did this to ensure that a fresh binary is built after setting gdb=1 in the configuration file.)

```
student@student-VirtualBox:~/kernel-1$ make clean
make[1]: Entering directory '/home/student/kernel-1/kernel'
make[1]: Leaving directory '/home/student/kernel-1/kernel'
make[1]: Entering directory '/home/student/kernel-1/user'
rm -f disk0.img lib/ld-weenix.so lib/libc.a lib/libc.so lib/libtest.a lib/libtes
t.so bin/ed.exec bin/ls.exec bin/sh.exec bin/uname.exec bin/hd.exec bin/stat.exe
c sbin/halt.exec sbin/init.exec usr/bin/args.exec usr/bin/hello.exec usr/bin/for
k-and-wait.exec usr/bin/kshell.exec usr/bin/segfault.exec usr/bin/spin.exec usr/
bin/eatmem.exec usr/bin/forkbomb.exec usr/bin/memtest.exec usr/bin/stress.exec u
sr/bin/vfstest.exec usr/bin/wc.exec usr/bin/forktest.exec usr/bin/eatinodes.exec
 usr/bin/pipetest.exec \
lib/libc/syscall.o lib/libc/strtol.o lib/libc/quad.o lib/libc/string.o lib/libc/
entry.o lib/libc/vsscanf.o lib/libc/scanf.o lib/libc/errno.o lib/libc/malloc.o l
ib/libc/stream.o lib/libc/rand.o lib/libc/vsnprintf.o lib/libc/printf.o lib/libt
est/test.o lib/ld-weenix/ldstart.o lib/ld-weenix/ldutil.o lib/ld-weenix/ldalloc.
o lib/ld-weenix/ldresolve.o lib/ld-weenix/ldnames.o lib/ld-weenix/ldreloc_i386.o
 lib/ld-weenix/bs_i386.o
make[1]: Leaving directory '/home/student/kernel-1/user'
student@student-VirtualBox:~/kernel-1$ make
make[1]: Entering directory '/home/student/kernel-1/kernel'
  Compiling "kernel/main/apic.c"...
  Compiling "kernel/main/pit.c"...
  Compiling "kernel/main/interrupt.c"...
  Compiling "kernel/main/acpi.c"...
```

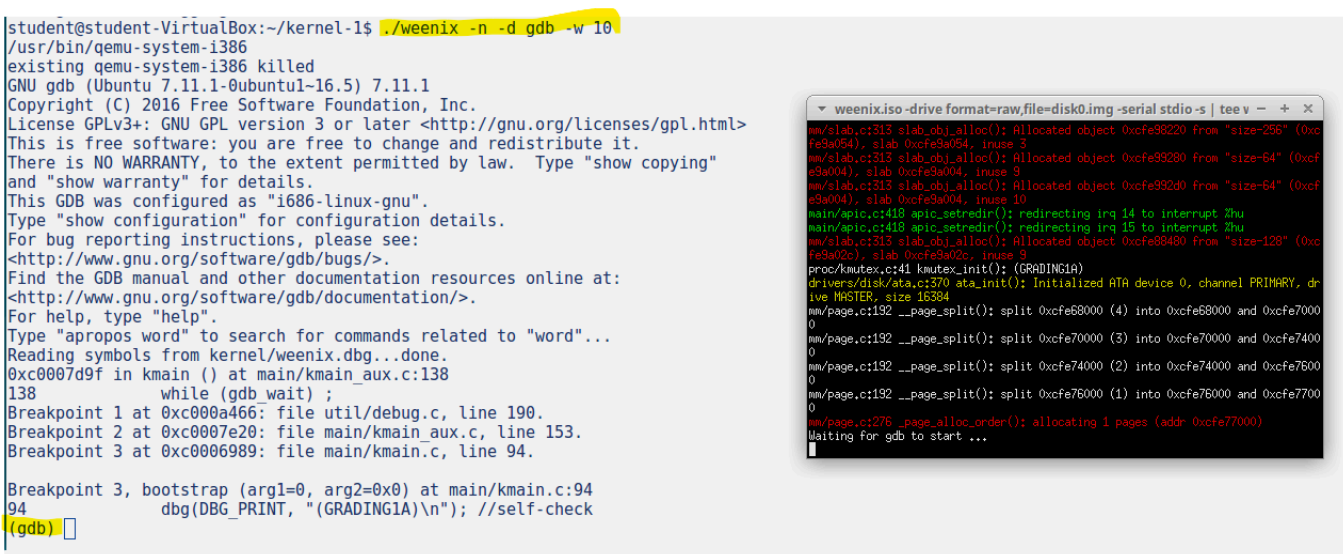
2. New binary is generated (weenix)

```
student@student-VirtualBox:~/kernel-1$ ls -la
total 1212
drwxrwxr-x  9 student student    4096 Mar 13 20:59 .
drwxr-xr-x 38 student student    4096 Mar 13 20:58 ..
-rw-rw-r--  1 student student     507 Mar 15  2025 commands.gdb
drwxrwxr-x  2 student student    4096 Mar 13  2025 compiler
-rw-rw-r--  1 student student    5714 Mar 13 20:58 Config.mk
drwxrwxr-x  3 student student    4096 Mar 13  2025 doc
-rwxrwxr-x  1 student student     120 Mar 13  2025 fsmaker
drwxrwxr-x  8 student student    4096 Mar 15  2025 .git
-rw-rw-r--  1 student student     843 Mar 13  2025 .gitlab-ci.yml
-rw-rw-r--  1 student student     741 Mar 13  2025 Global.mk
-rw-rw-r--  1 student student     121 Mar 13  2025 init.gdb
-rw-rw-r--  1 student student      85 Mar 15  2025 inittmp.gdb
drwxrwxr-x 15 student student    4096 Mar 13 20:59 kernel
-rw-rw-r--  1 student student    1892 Mar 13  2025 Makefile
-rw-rw-r--  1 student student      23 Mar 13  2025 memcheck.gdb
-rw-rw-r--  1 student student    4851 Mar 15  2025 procs-README.txt
-rw-rw-r--  1 student student   16291 Mar 15  2025 procs-submit.tar.gz
drwxrwxr-x  3 student student    4096 Mar 13  2025 python
-rw-rw-r--  1 student student     684 Mar 13  2025 README
-rw-rw-r--  1 student student    5940 Mar 13  2025 README.gdb
drwxrwxr-x  3 student student    4096 Mar 13  2025 tools
drwxrwxr-x  9 student student    4096 Mar 13 20:59 user
-rw-rw-r--  1 student student    4823 Mar 13  2025 Vagrantfile
-rw-rw-r--  1 student student      76 Mar 13  2025 vfs-README.txt
-rw-rw-r--  1 student student      76 Mar 13  2025 vm-README.txt
-rwxrwxr-x  1 student student    5847 Mar 13  2025 weenix
-rw-rw-r--  1 student student 1096559 Mar 15  2025 weenix.log
-rwxrwxr-x  1 student student     952 Mar 13  2025 weenix-tmux-debug
student@student-VirtualBox:~/kernel-1$
```

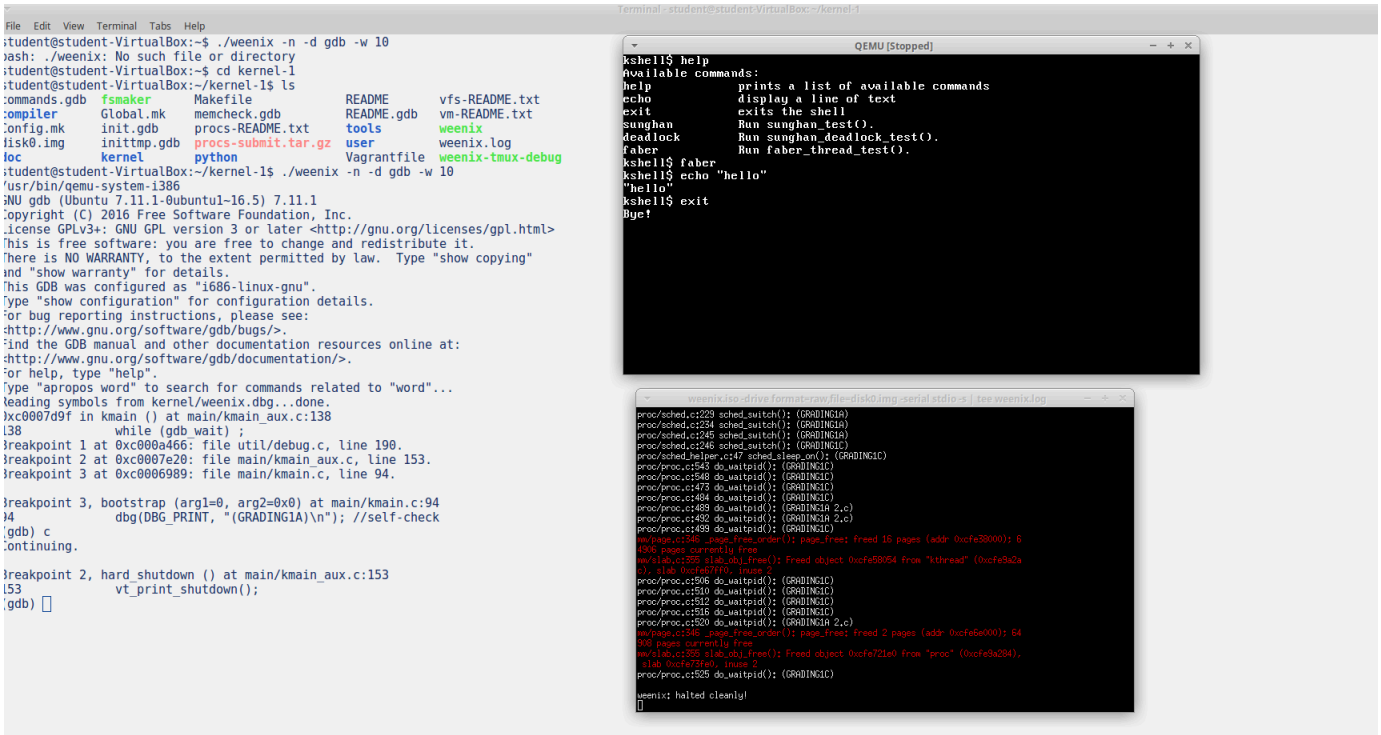
3. Weenix Kernel: Qemu



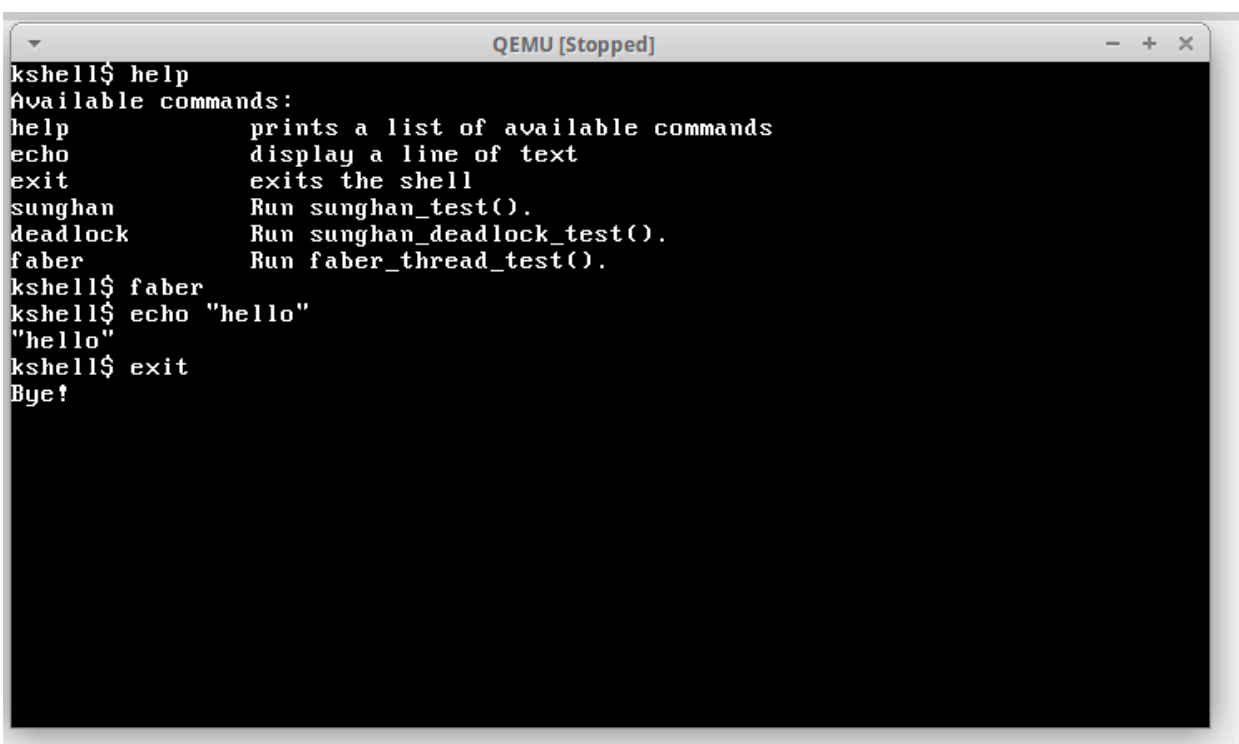
4. Weenix kernel start with logs, gdb



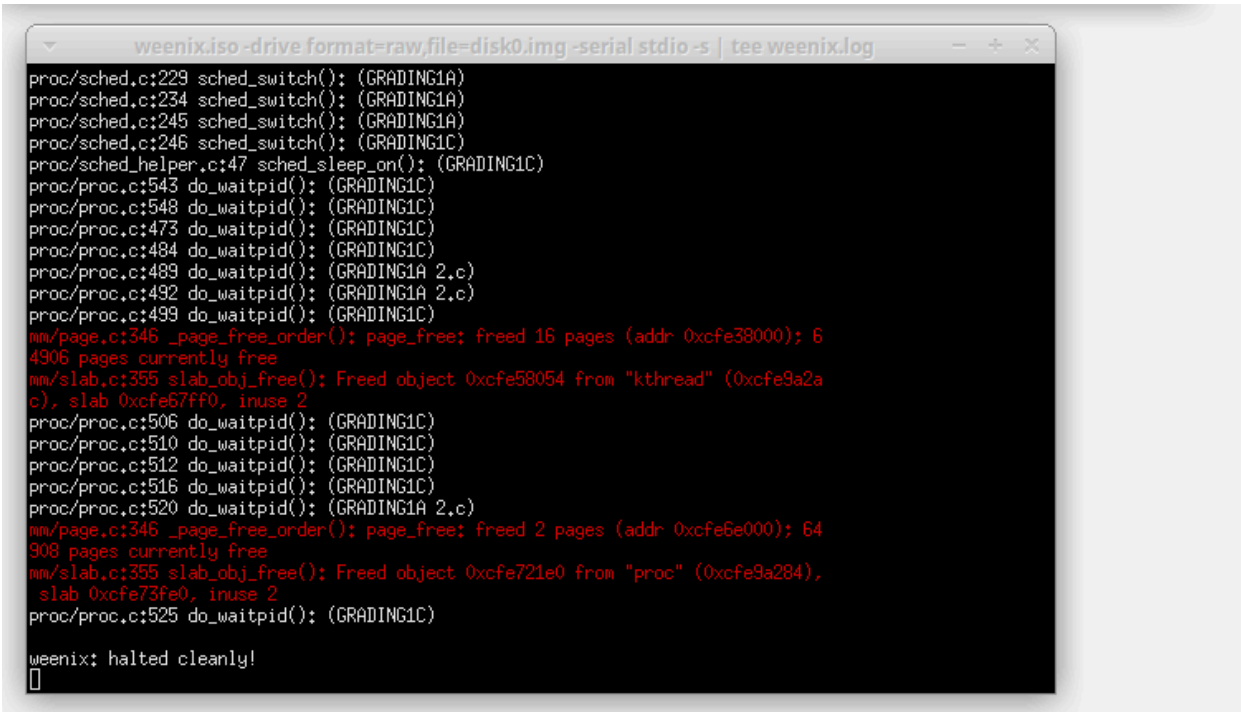
5. Kshell, and running faber tests



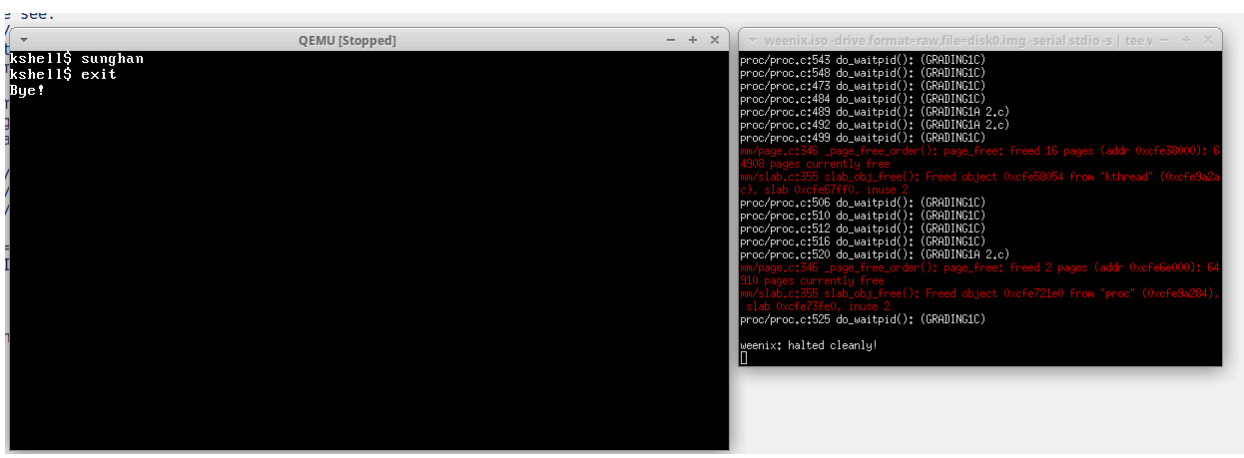
6. Zooming in on the Qemu terminal



7. Zooming in on the logs



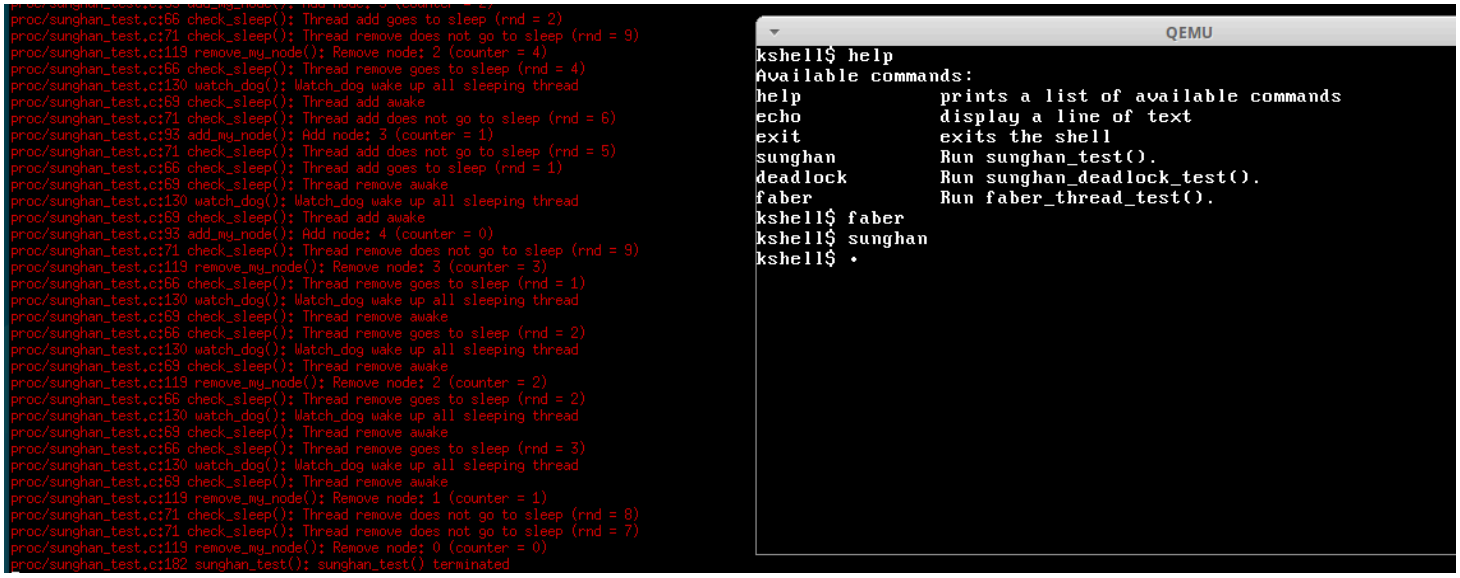
8. Sunghan test



Config file with DBG=error,test (instead of DBG=all), to show specific faber tests and sunghan logs:

9. Faber tests

```
proc/faber_test.c:328 faber_thread_test():
>>> Start running faber_thread_test()...
proc/faber_test.c:331 faber_thread_test(): (C.1) start
proc/faber_test.c:332 faber_thread_test(): waitpid any test
proc/faber_test.c:102 wait_for_any(): child (4) exited: 23
proc/faber_test.c:336 faber_thread_test(): waitpid test
proc/faber_test.c:90 wait_for_proc(): waitpid test (5) exited: 32
proc/faber_test.c:342 faber_thread_test(): kthread exit test
proc/faber_test.c:90 wait_for_proc(): kthread exit test (6) exited: 0
proc/faber_test.c:346 faber_thread_test(): many test
proc/faber_test.c:102 wait_for_any(): child (7) exited: 0
proc/faber_test.c:102 wait_for_any(): child (8) exited: 1
proc/faber_test.c:102 wait_for_any(): child (9) exited: 2
proc/faber_test.c:102 wait_for_any(): child (10) exited: 3
proc/faber_test.c:102 wait_for_any(): child (11) exited: 4
proc/faber_test.c:102 wait_for_any(): child (12) exited: 5
proc/faber_test.c:102 wait_for_any(): child (13) exited: 6
proc/faber_test.c:102 wait_for_any(): child (14) exited: 7
proc/faber_test.c:102 wait_for_any(): child (15) exited: 8
proc/faber_test.c:102 wait_for_any(): child (16) exited: 9
proc/faber_test.c:350 faber_thread_test(): (C.1) done
proc/faber_test.c:355 faber_thread_test(): (C.2) start
proc/faber_test.c:356 faber_thread_test(): Context switch test
proc/faber_test.c:90 wait_for_proc(): Context switch (17) exited: 1
proc/faber_test.c:359 faber_thread_test(): (C.2) done
proc/faber_test.c:366 faber_thread_test(): (C.3) start
proc/faber_test.c:367 faber_thread_test(): wake me test
proc/faber_test.c:90 wait_for_proc(): wake me test (18) exited: 0
proc/faber_test.c:376 faber_thread_test(): broadcast me test
proc/faber_test.c:102 wait_for_any(): child (19) exited: 0
proc/faber_test.c:102 wait_for_any(): child (20) exited: 0
proc/faber_test.c:102 wait_for_any(): child (21) exited: 0
proc/faber_test.c:102 wait_for_any(): child (22) exited: 0
proc/faber_test.c:102 wait_for_any(): child (23) exited: 0
proc/faber_test.c:102 wait_for_any(): child (24) exited: 0
proc/faber_test.c:102 wait_for_any(): child (25) exited: 0
proc/faber_test.c:102 wait_for_any(): child (26) exited: 0
proc/faber_test.c:102 wait_for_any(): child (27) exited: 0
proc/faber_test.c:102 wait_for_any(): child (28) exited: 0
proc/faber_test.c:384 faber_thread_test(): (C.3) done
proc/faber_test.c:389 faber_thread_test(): (C.4) start
proc/faber_test.c:390 faber_thread_test(): wake me uncancellable test
proc/faber_test.c:90 wait_for_proc(): wake me uncancellable test (29) exited: 0
proc/faber_test.c:399 faber_thread_test(): broadcast me uncancellable test
proc/faber_test.c:102 wait_for_any(): child (30) exited: 0
proc/faber_test.c:102 wait_for_any(): child (31) exited: 0
proc/faber_test.c:102 wait_for_any(): child (32) exited: 0
proc/faber_test.c:102 wait_for_any(): child (33) exited: 0
proc/faber_test.c:102 wait_for_any(): child (34) exited: 0
proc/faber_test.c:102 wait_for_any(): child (35) exited: 0
proc/faber_test.c:102 wait_for_any(): child (36) exited: 0
proc/faber_test.c:102 wait_for_any(): child (37) exited: 0
proc/faber_test.c:102 wait_for_any(): child (38) exited: 0
proc/faber_test.c:102 wait_for_any(): child (39) exited: 0
proc/faber_test.c:408 faber_thread_test(): (C.4) done
```

```
proc/faber_test.c:414 faber_thread_test(): cancel me test
proc/faber_test.c:90 wait_for_proc(): cancel me test (40) exited: 0
proc/faber_test.c:422 faber_thread_test(): prior cancel me test
proc/faber_test.c:90 wait_for_proc(): prior cancel me test (41) exited: 0
proc/faber_test.c:429 faber_thread_test(): cancel me head test
proc/faber_test.c:102 wait_for_any(): child (42) exited: 0
proc/faber_test.c:102 wait_for_any(): child (43) exited: 0
proc/faber_test.c:438 faber_thread_test(): cancel me tail test
proc/faber_test.c:102 wait_for_any(): child (44) exited: 0
proc/faber_test.c:102 wait_for_any(): child (45) exited: 0
proc/faber_test.c:446 faber_thread_test(): (C.5) done
proc/faber_test.c:451 faber_thread_test(): (C.6) start
proc/faber_test.c:452 faber_thread_test(): Reparenting test
proc/faber_test.c:102 wait_for_any(): child (46) exited: 0
proc/faber_test.c:458 faber_thread_test(): Reparenting stress test
proc/faber_test.c:102 wait_for_any(): child (48) exited: 0
proc/faber_test.c:465 faber_thread_test(): (C.6) done
proc/faber_test.c:472 faber_thread_test(): (C.7) start
proc/faber_test.c:473 faber_thread_test(): show race test
proc/faber_test.c:102 wait_for_any(): child (68) exited: 1
proc/faber_test.c:102 wait_for_any(): child (69) exited: 1
proc/faber_test.c:102 wait_for_any(): child (70) exited: 1
proc/faber_test.c:102 wait_for_any(): child (71) exited: 1
proc/faber_test.c:102 wait_for_any(): child (72) exited: 1
proc/faber_test.c:102 wait_for_any(): child (73) exited: 1
proc/faber_test.c:102 wait_for_any(): child (74) exited: 1
proc/faber_test.c:102 wait_for_any(): child (75) exited: 1
proc/faber_test.c:102 wait_for_any(): child (76) exited: 1
proc/faber_test.c:102 wait_for_any(): child (77) exited: 1
proc/faber_test.c:479 faber_thread_test(): fix race test
proc/faber_test.c:102 wait_for_any(): child (78) exited: 1
proc/faber_test.c:102 wait_for_any(): child (79) exited: 2
proc/faber_test.c:102 wait_for_any(): child (80) exited: 3
proc/faber_test.c:102 wait_for_any(): child (81) exited: 4
proc/faber_test.c:102 wait_for_any(): child (82) exited: 5
proc/faber_test.c:102 wait_for_any(): child (83) exited: 6
proc/faber_test.c:102 wait_for_any(): child (84) exited: 7
proc/faber_test.c:102 wait_for_any(): child (85) exited: 8
proc/faber_test.c:102 wait_for_any(): child (86) exited: 9
proc/faber_test.c:102 wait_for_any(): child (87) exited: 10
proc/faber_test.c:485 faber_thread_test(): fix race test w/cancel
proc/faber_test.c:102 wait_for_any(): child (89) exited: 0
proc/faber_test.c:102 wait_for_any(): child (91) exited: 0
proc/faber_test.c:102 wait_for_any(): child (93) exited: 0
proc/faber_test.c:102 wait_for_any(): child (95) exited: 0
proc/faber_test.c:102 wait_for_any(): child (97) exited: 0
proc/faber_test.c:102 wait_for_any(): child (88) exited: 1
proc/faber_test.c:102 wait_for_any(): child (90) exited: 2
proc/faber_test.c:102 wait_for_any(): child (92) exited: 3
proc/faber_test.c:102 wait_for_any(): child (94) exited: 4
proc/faber_test.c:102 wait_for_any(): child (96) exited: 5
proc/faber_test.c:496 faber_thread_test(): (C.7) done
proc/faber_test.c:501 faber_thread_test(): (C.8) start
proc/faber_test.c:502 faber_thread_test(): kill child procs test
proc/faber_test.c:102 wait_for_any(): child (98) exited: 0
proc/faber_test.c:102 wait_for_any(): child (99) exited: 0
proc/faber_test.c:102 wait_for_any(): child (100) exited: 0
proc/faber_test.c:102 wait_for_any(): child (101) exited: 0
proc/faber_test.c:102 wait_for_any(): child (102) exited: 0
proc/faber_test.c:102 wait_for_any(): child (103) exited: 0
proc/faber_test.c:102 wait_for_any(): child (104) exited: 0
proc/faber_test.c:102 wait_for_any(): child (105) exited: 0
proc/faber_test.c:102 wait_for_any(): child (106) exited: 0
proc/faber_test.c:102 wait_for_any(): child (107) exited: 0
proc/faber_test.c:511 faber_thread_test(): (C.8) done
proc/faber_test.c:516 faber_thread_test(): (C.9) start (no end)
proc/faber_test.c:517 faber_thread_test(): proc kill all test
```

10. Sunghan test with logs